

Literature Review and Theoretical Foundations of Stacked Generalization for Multi-Modal Skin Cancer Classification

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Abstract—This literature review establishes the theoretical foundations underlying the Phase 3 stacking architecture used in our ISIC 2024 skin cancer classification pipeline. While modern ensemble systems frequently employ stacking as an empirical engineering technique, the methodology originates from the formal stacked generalization framework introduced by Wolpert. This review connects our engineering decisions — including out-of-fold prediction generation, heterogeneous ensemble diversity, rank-space feature transformation, and logistic regression meta-learning — to the original theoretical principles of stacked generalization. We further relate these concepts to modern developments in gradient boosting, transfer learning, calibration theory, and cross-modal ensemble learning under extreme class imbalance. The review demonstrates that our Phase 3 architecture is not merely an ensemble heuristic, but a theoretically grounded meta-learning system designed to estimate and correct the generalization biases of heterogeneous base learners.

I. INTRODUCTION

Skin cancer classification presents a uniquely difficult machine learning problem due to:

- extreme class imbalance,
- high intra-class variability,
- patient-level correlation,
- heterogeneous feature modalities,
- and calibration-sensitive evaluation metrics.

The ISIC 2024 SLICE-3D challenge introduces a particularly demanding setting containing:

- structured clinical metadata,
- high-resolution dermoscopic images,
- and severe imbalance ($\approx 1 : 1020$ malignant-to-benign ratio).

Traditional single-model approaches struggle to capture the complementary information distributed across these modalities. Consequently, ensemble learning and meta-learning techniques have become central to state-of-the-art medical imaging systems.

Our research pipeline consists of:

- 1) tabular gradient-boosted tree ensembles,
- 2) deep image classification backbones,
- 3) and a stacking-based meta-learner.

The purpose of this literature review is to formally connect the engineering architecture of our Phase 3 stacking system to the theoretical foundations of stacked generalization introduced by Wolpert [1]. Specifically, we show that our implementation follows the core principles of:

- bias estimation,
- out-of-sample meta-learning,
- heterogeneous generalizer fusion,
- and level-1 error correction.

II. FOUNDATIONS OF STACKED GENERALIZATION

A. Original Theory of Stacked Generalization

Stacked generalization was formally introduced by David H. Wolpert in 1992 as a method for minimizing generalization error through a second-level learning process [1]. Unlike winner-takes-all approaches such as cross-validation model selection, stacked generalization learns how to combine multiple base learners through a higher-order meta-learner.

The central idea is that each base learner possesses systematic biases:

$$g_i(x) = f(x) + \epsilon_i(x) \quad (1)$$

where:

- $f(x)$ is the true underlying function,
- $g_i(x)$ is the prediction of the i -th model,
- $\epsilon_i(x)$ represents the model-specific error structure.

Instead of selecting a single “best” model, stacked generalization trains a second-level learner:

$$M(g_1(x), g_2(x), \dots, g_n(x)) \quad (2)$$

to estimate the optimal combination of base predictions.

Wolpert argued that ensemble improvement emerges only when:

- different models make different errors,
- and the meta-learner can statistically infer these error patterns.

This principle directly motivates the heterogeneous architecture diversity used in our Phase 3 system.

B. Level-0 and Level-1 Learning

Wolpert defines:

- **Level-0 learners:** original base models,
- **Level-1 learner:** meta-model trained on predictions of level-0 learners.

In our implementation:

Our Phase 3 architecture therefore directly instantiates Wolpert’s original stacked generalization framework.

TABLE I
MAPPING BETWEEN STACKED GENERALIZATION THEORY AND OUR SYSTEM

Stacking Theory	Our Implementation
Level-0 generalizers	LightGBM, XGBoost, CatBoost, EfficientNetV2, EVA02, ConvNeXtV2, SwinV2
Level-1 feature space	Rank-transformed OOF predictions
Level-1 generalizer	Logistic Regression
Cross-validation partitions	StratifiedGroupKFold
Bias estimation	Learned modality weighting
Generalization optimization	pAUC maximization

III. OUT-OF-FOLD PREDICTIONS AND GENERALIZATION BIAS

A. Importance of Out-of-Sample Meta-Learning

One of the most critical theoretical principles of stacked generalization is that the meta-learner must train on out-of-sample predictions rather than training predictions.

If a base model predicts on samples it was trained on:

$$P(\hat{y}_{train} \approx y) \gg P(\hat{y}_{test} \approx y) \quad (3)$$

then the predictions become optimistically biased estimates of true generalization performance.

Consequently, the level-1 learner would learn:

- memorization behavior,
- rather than true error structure.

To avoid this, Wolpert proposed partition-based learning using held-out subsets.

Our implementation follows this principle exactly through:

- identical StratifiedGroupKFold partitions,
- patient-wise leakage prevention,
- and out-of-fold prediction alignment.

This design ensures that the meta-learner observes only unbiased approximations of generalization behavior.

B. Patient-Level Leakage Prevention

Medical datasets frequently contain correlated samples from the same patient. Random splitting causes:

$$x_i^{train} \sim x_j^{val} \quad (4)$$

for lesions belonging to the same patient.

This introduces severe leakage because:

- the model can memorize patient-specific distributions,
- inflating validation performance.

Our use of:

$$\text{StratifiedGroupKFold}(\text{groups}=\text{patient_id}) \quad (5)$$

ensures that no patient appears in both training and validation folds.

This preserves the statistical validity of the level-1 training space.

IV. ARCHITECTURAL DIVERSITY AND ENSEMBLE THEORY

A. Bias Diversity as the Core Requirement

Ensemble learning improves performance only when constituent models exhibit diverse error manifolds.

If:

$$\epsilon_1(x) \approx \epsilon_2(x) \quad (6)$$

then stacking provides minimal gain because the models fail identically.

Conversely, when:

$$\text{Cov}(\epsilon_i, \epsilon_j) \rightarrow 0 \quad (7)$$

the meta-learner can exploit complementary strengths.

This principle strongly influenced our architecture selection strategy.

B. Heterogeneous Base Learners

Our system intentionally combines:

- gradient-boosted decision trees,
- convolutional neural networks,
- vision transformers,
- hybrid hierarchical transformers,
- and tabular-conditioned visual models.

Each architecture possesses distinct inductive biases:

TABLE II
ARCHITECTURAL BIAS DIVERSITY

Model Type	Primary Bias
GBDTs	Tabular feature interactions
CNNs	Local spatial texture priors
ViTs	Global self-attention relationships
SwinV2	Hierarchical local-global reasoning
CatBoost	Ordered categorical boosting

This diversity is essential for stacked generalization to improve over any single learner.

V. RANK TRANSFORMATION AND META-FEATURE GEOMETRY

A. Motivation for Rank-Space Learning

The outputs of heterogeneous models often exist on incompatible scales:

- calibrated probabilities,
- sigmoid outputs,
- boosted tree scores,
- or poorly calibrated logits.

Directly combining these values introduces scale dominance:

$$\|p_i\| \gg \|p_j\| \quad (8)$$

causing the meta-learner to overweight certain modalities.

To address this, our system applies percentile rank transformation:

$$r_m(i) = \frac{\text{rank}(\hat{p}_m(i))}{N} \quad (9)$$

This creates:

- scale invariance,
- monotonicity preservation,
- robustness to calibration mismatch,
- and stable ranking geometry.

B. Connection to AUC Optimization

The ISIC challenge uses partial AUC (pAUC), which depends primarily on:

- prediction ordering,
- rather than absolute probability magnitude.

Rank transformation therefore aligns naturally with the evaluation objective.

Because AUC is invariant under monotonic transformations:

$$AUC(f(x)) = AUC(g(f(x))) \quad (10)$$

for monotonic g , rank-space learning preserves discriminative structure while stabilizing heterogeneous prediction distributions.

VI. LOGISTIC REGRESSION AS A META-LEARNER

A. Why Linear Meta-Learning Generalizes Better

Our experiments showed that logistic regression outperformed tree-based stackers despite the latter possessing higher representational capacity.

This behavior is theoretically expected.

The Phase 3 meta-space contains:

- only 5 features,
- and approximately 393 positive samples.

Under such low-information conditions, highly expressive learners:

- overfit noise,
- memorize fold-specific artifacts,
- and fail to generalize.

Logistic regression instead learns:

$$\hat{y} = \sigma(w_1x_1 + w_2x_2 + \dots + w_nx_n) \quad (11)$$

This creates:

- smooth decision boundaries,
- low variance,
- and stable modality weighting.

B. Meta-Learning Interpretation

The Phase 3 stacker can therefore be interpreted as:

- a shallow meta-learning system,
- estimating the reliability of heterogeneous modalities.

Rather than simply averaging predictions, the stacker learns:

- when image models are trustworthy,
- when tabular priors dominate,
- and how cross-modal disagreement correlates with malignancy.

This behavior directly matches Wolpert’s original formulation of stacked generalization as:

“learning how generalizers generalize.”

VII. CROSS-MODAL LEARNING IN MEDICAL AI

Modern medical AI increasingly relies on multi-modal learning because:

- images encode morphology,
- while metadata encodes clinical priors.

In skin cancer classification:

- dermoscopic images capture border irregularity,
- texture,
- pigmentation,
- and asymmetry,

whereas tabular features encode:

- age,
- anatomical location,
- lesion evolution,
- and patient-relative abnormality.

Our stacking architecture combines these orthogonal information sources within a unified meta-learning framework.

The resulting performance gain supports the hypothesis that:

$$I(X_{image}; Y) \not\subseteq I(X_{tabular}; Y) \quad (12)$$

meaning the modalities contain complementary predictive information.

VIII. LIMITATIONS AND FUTURE DIRECTIONS

Although effective, our current stacking system possesses several limitations.

A. Linear Fusion Constraints

Logistic regression cannot model:

- nonlinear modality interactions,
- uncertainty-aware routing,
- or dynamic confidence weighting.

B. Absence of Uncertainty Modeling

The stacker currently uses:

- only prediction ranks,

while ignoring:

- entropy,
- calibration uncertainty,
- disagreement statistics,
- and Bayesian confidence estimates.

Future work could explore:

- Bayesian stacking,
- mixture-of-experts routing,
- uncertainty-aware fusion,
- or transformer-based meta-learners.

C. Single-Level Stacking

Our architecture employs only one meta-learning layer. Multi-level stacking architectures may further improve:

- hierarchical error correction,
- modality specialization,
- and robustness under distribution shift.

IX. CONCLUSION

This review demonstrates that our Phase 3 system is firmly grounded in the theoretical framework of stacked generalization introduced by Wolpert. Our engineering choices — including:

- out-of-fold meta-learning,
- patient-grouped cross-validation,
- heterogeneous architectural diversity,
- rank-space transformation,
- and low-variance logistic meta-learning

all directly follow established ensemble learning principles.

The resulting system functions not merely as an ensemble heuristic, but as a theoretically motivated meta-learning architecture capable of learning and correcting the generalization biases of heterogeneous multi-modal learners.

REFERENCES

- [1] D. H. Wolpert, “Stacked Generalization,” *Neural Networks*, vol. 5, no. 2, pp. 241–259, 1992.